COMPUTER NETWORK

What is a Computer Network?

A group of two or more computers connected to each other by:

a *cable*,

telephone lines, or

through *wireless communication*.

Local Area Network (LAN)

An interconnection of computers, terminals, and other devices at a single geographical site, for example, within the same adjacent buildings.

Wide Area Network

A connection of computers in different locations either national or international.

It is designed to cover a wider area than a local area network.





For a computer to communicate with another computer over the telephone network, a MODEM (short for **Mo**dulator **Dem**odulator) is needed.

Changes data from digital (machine code) to analog form so that it can be carried over the telephone lines.

At the receiving end another modem changes that data from analogue back to digital form for the computer to read and understand.





WHAT IS THE INTERNET?

A network made up of hundred of thousand of computers all over the world, with millions of users connected electronically

What can you do with the Internet?

- Information Browsing
- File Transfer
- Use of other computers
- international debates or discussion groups
- Electronic mail





WHAT IS THE WEB?

The Web is the part or service of the Internet that uses text, images, sound, and video.

It links documents together

This linking is sometimes called hypertext.

The **World Wide Web** is a collection of electronic documents that are linked together like a spider web.

These documents are stored on computers called **servers** located around the world





What information can I find on the Web?

- You can find something about just any subject on the Web.
- All kinds of people and organizations put up information on the Web
- Things to be aware of in order to use this tool effectively
- Not everything on the Web is accurate or authoritative. Evaluate the source of the information you find.
- They are things on the Web that you may find offensive.
- Not everything can be found on the Web.





What Is the Web Made of?

The Web consists of:

- Your **personal computer**
- **Web browser software** to access the Web
- A connection to an **Internet service provider** (ISP)
- **Servers** to host the data
- **Routers** and **switches** to direct the flow of data





What is a Web Browser?

- Software that provides users with a graphical interface for connecting to and navigating through the World Wide Web.
- Used to access the World Wide Web.
- It retrieves data from remote web servers and displays a web page.





Types of Browsers

Internet Explorer and *Navigator* are the two most popular browser programs currently in use.

These two browsers work approximately the same way. Some terminology is different.

Both are being constantly upgraded to make them more competitive with one another.





STARTING MICROSOFT INTERNET EXPLORER

To start Microsoft Explorer:

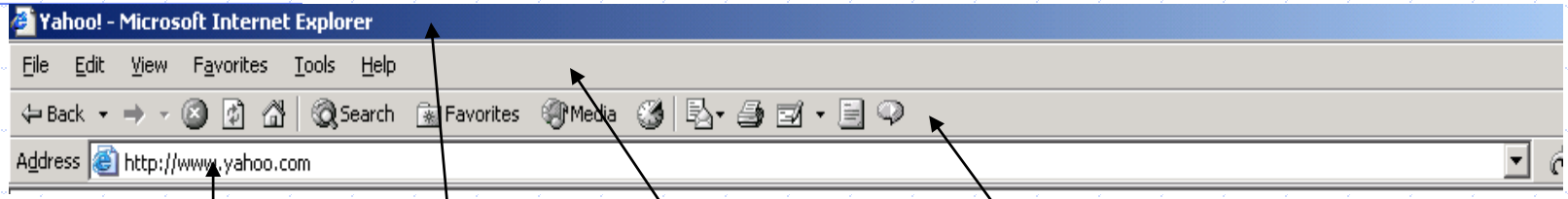
- Choose the Internet Explorer command from the Start menu

Or

- Double-click on the Internet Explorer icon 

on the desktop.

Anatomy of Microsoft Internet Explorer



Address bar Title bar Menu bar Standard tool bar

Anatomy of Microsoft Internet Explorer…

- Title bar
- Menu bar
- Standard toolbar
- Address bar
- Internet Explorer icon
- Scroll boxes and scroll arrows
- Status bar





Address bar displays:

- the Web address (URL) of the page you are currently viewing, or
- the location of the currently open file.

You can use it to enter the address or location of a page of file you want to open.

Internet Explorer icon

- Shows when a Web page or other file is being loaded into the browser window.
- As the page or file loads, the icon is animated.

Status bar

It displays the address or location of the page of file being opened, along with progress information.

Working with:

1. The Status bar
2. Text boxes
3. Toolbars
4. ScreenTips





What is a Web Page?

A **web page** is an electronic document written in a computer language called **HTML** (Hypertext Markup Language).

Web pages are stored on **web servers**

They can contain text, graphics, video, animation, links to other pages, links to other files, and sound





Uniform Resource Locators (URLs)

- Uniform Resource Locators (URLs) are the method by which all resources on the Web are addressed

Presentation of information in a URL:

Type_of_service://host_name_and_address:port_number/directory_path/resource_name

Example:

http://www.soc.staffs.ac.uk/news/news.html





http://www.soc.staffs.ac.uk/news/news.html

http – Use Hypertext Transfer Protocol (HTTP) to access this resource

www.soc.staffs.ac.uk - The name and address of the host computer – (Domain name)

www – A World Wide Web server at

soc (the School of Computing) at

staffs (Stafford University), which is an

academic site in the **uk**





news/news.html – resource name

news.html - a Web document stored in the **news** directory

Using URLs

A URL indicates where the web page is stored on the Internet.

Tip: URLs almost use forward slashes (/).





To enter a URL directly:



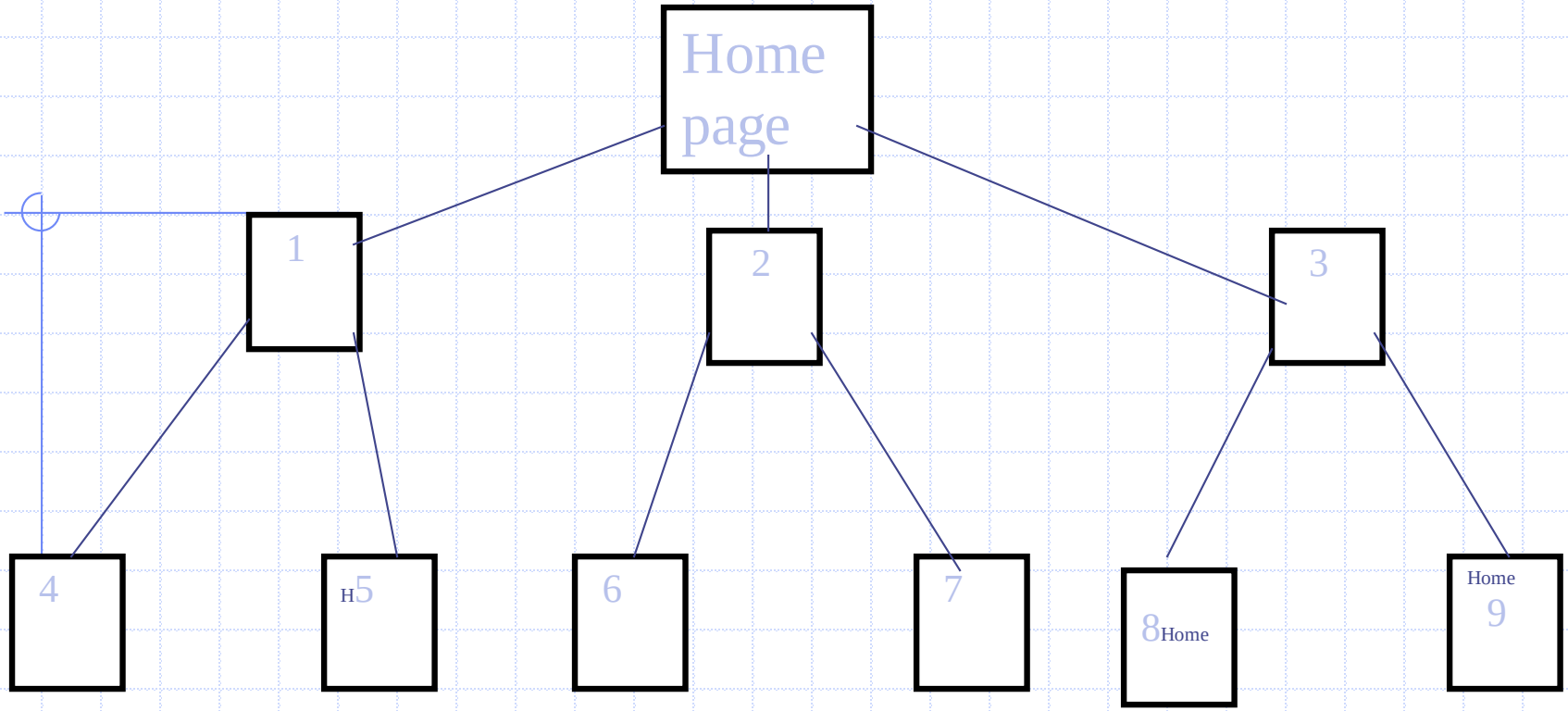
- Click in the **Address** bar
- Type the new URL
- Press **[Enter]**.

Websites

A **website** is one or more web pages that relate to a common theme, such as a person, business, organization, or a subject, such as sports.






Home page
1
2
3
4
H5
6
7
8Home
Home
9

Navigating the Web

Three main ways to move between web pages or websites:

- Clicking a **text hyperlink**.
- Clicking a **hyperlinked graphic image**
- Typing the **URL of a web page** in the **location box** and pressing the **Enter** key.





Hyperlinks

Hyperlinks usually connect you to one of the following:

- A different part of the same page
- A different page, either at the same site or somewhere else on the Web.
- A file that you can open or download to your computer.





- Coloured and/or underlined text (the default colour for a hyperlink is usually blue).
- When you position the mouse pointer over a hyperlink, the pointer changes to a hand: 
- A URL appears in the **status bar** at the lower left of your web browser.

Navigation Buttons on The Toolbar



Overview of searching

Search engines provide a convenient and easy way to finding some Web resources.

These search engines are found at sites like

Excite

Yahoo

AltaVista

Infoseek





How Search Engines Work

1. Traverse the Web, retrieving and indexing Web pages. Start with the sites they know
2. Seek out all the links embedded in those pages
3. Follow them, then repeat the process for all the new sites they come across.
4. Compile enormous databases of information, addresses and links.

These databases are accessed when someone submits an enquiry.





Some popular search engines include:

Excite (http://www.excite.com)

HotBot (http://www.hotbot.com)

Infoseek (http://www.infoseek.com)

Northern Light (http://www.northernlight.com)





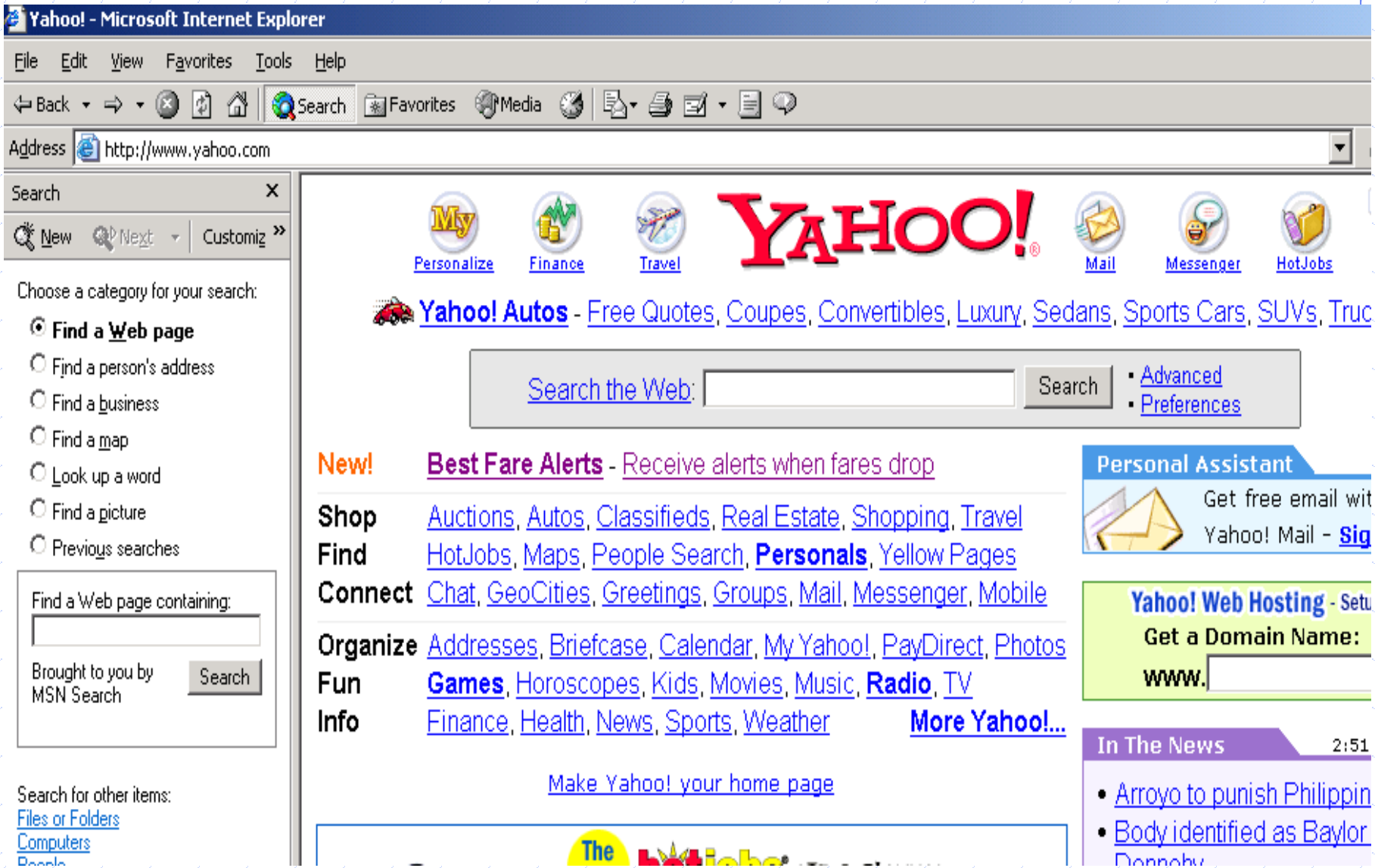
Using Yahoo

- Yahoo classifies all the Web pages in its database into a hierarchy of categories.
- You can search for all Web pages in the Yahoo database by entering key words or by selecting the appropriate categories.
- To access Yahoo, navigate to the following URL:

http://www.yahoo.com






Yahoo! - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Search Favorites Media
Address http://www.yahoo.com
Search
New Next Customiz
Choose a category for your search:
Find a Web page
Find a person's address
Find a business
Find a map
Look up a word
Find a picture
Previous searches
Find a Web page containing:
Brought to you by MSN Search
Search
Search for other items:
Files or Folders
Computers
My Personalize
Finance
Travel
YAHOO!
Mail
Messenger
HotJobs
Yahoo! Autos - Free Quotes, Coupes, Convertibles, Luxury, Sedans, Sports Cars, SUVs, Truc
Search the Web:
Search
Advanced
Preferences
New! Best Fare Alerts - Receive alerts when fares drop
Shop Auctions, Autos, Classifieds, Real Estate, Shopping, Travel
Find HotJobs, Maps, People Search, Personals, Yellow Pages
Connect Chat, GeoCities, Greetings, Groups, Mail, Messenger, Mobile
Organize Addresses, Briefcase, Calendar, My Yahoo!, PayDirect, Photos
Fun Games, Horoscopes, Kids, Movies, Music, Radio, TV
Info Finance, Health, News, Sports, Weather
More Yahoo!...
Make Yahoo! your home page
Personal Assistant
Get free email wit
Yahoo! Mail - Sig
Yahoo! Web Hosting - Setu
Get a Domain Name:
www.
In The News 2:51
Arroyo to punish Philippin
Body identified as Baylor

Searching Yahoo for key words or phrases:

To search the Web pages containing a particular word or phrase, type your search terms in the text box on the form, then click on the **Search** button.

Searching Yahoo by Category:

- You can search through a hierarchy of categories to find web pages you are interest in.
- In the main body of the Yahoo home page, click on the general category in which you are interested.

Search Syntax

- Capitalising Names and Titles
- Using commas (,) to separate Names and Titles
- Using Double Quotation marks (“) or Hyphens (-) to indicate Phrases
- Using a plus sign (+) in front of a word that must appear
- Using a minus sign (-) in front of a word that must not appear
- Using an Asterisk (*) to search for partial matches





COPYRIGHT ON THE INTERNET

- It is extraordinarily easy to copy things on the Internet.
- If something is physically easy to copy it does not mean that it is legal to copy it.
- Absence of a copyright notice does not mean it is okay to copy it.
- When using information taken from the Internet, always reference the Web site, the title of the information and the author.
- It is also important to reference the date of the information on the Internet.





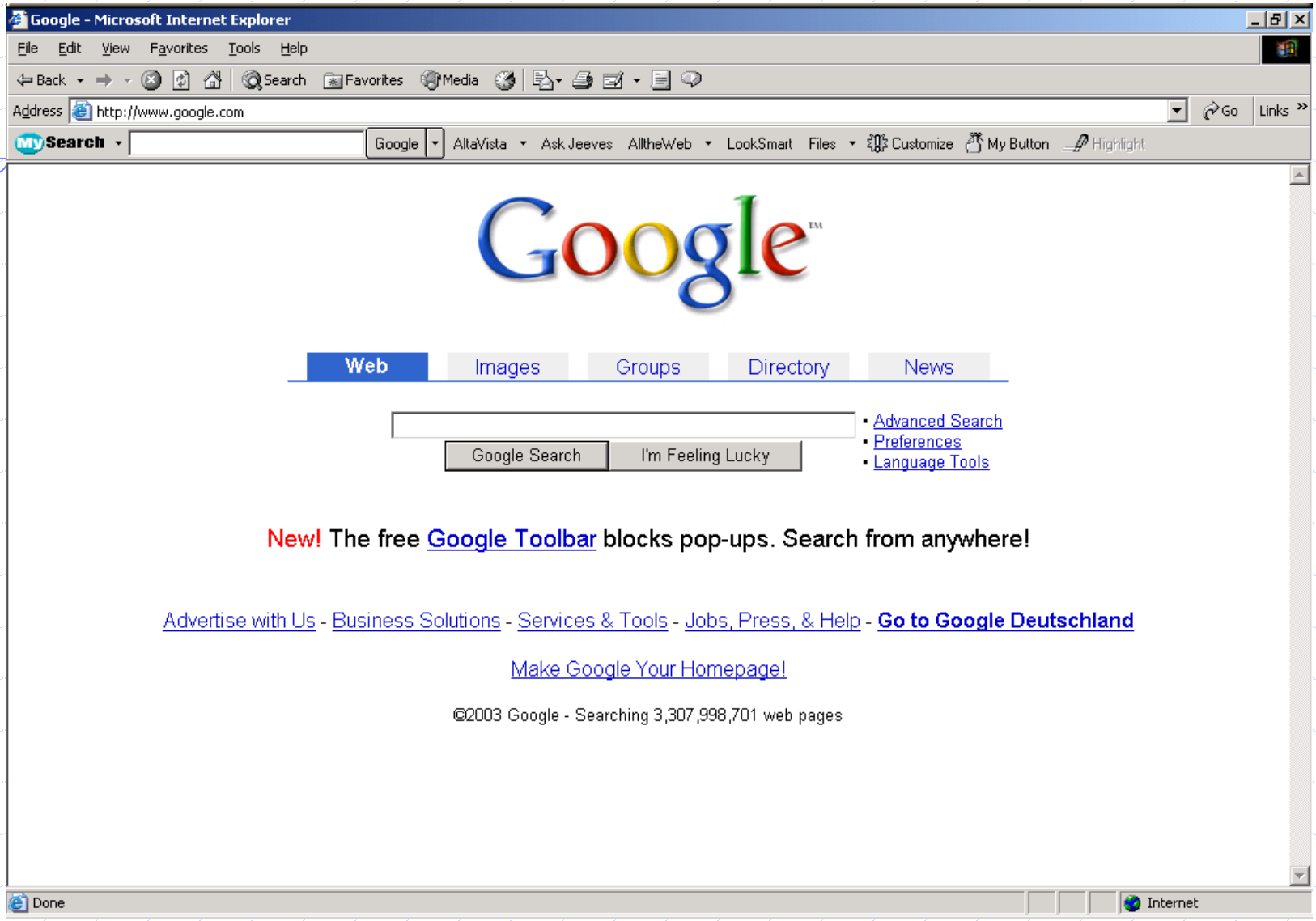
Example:

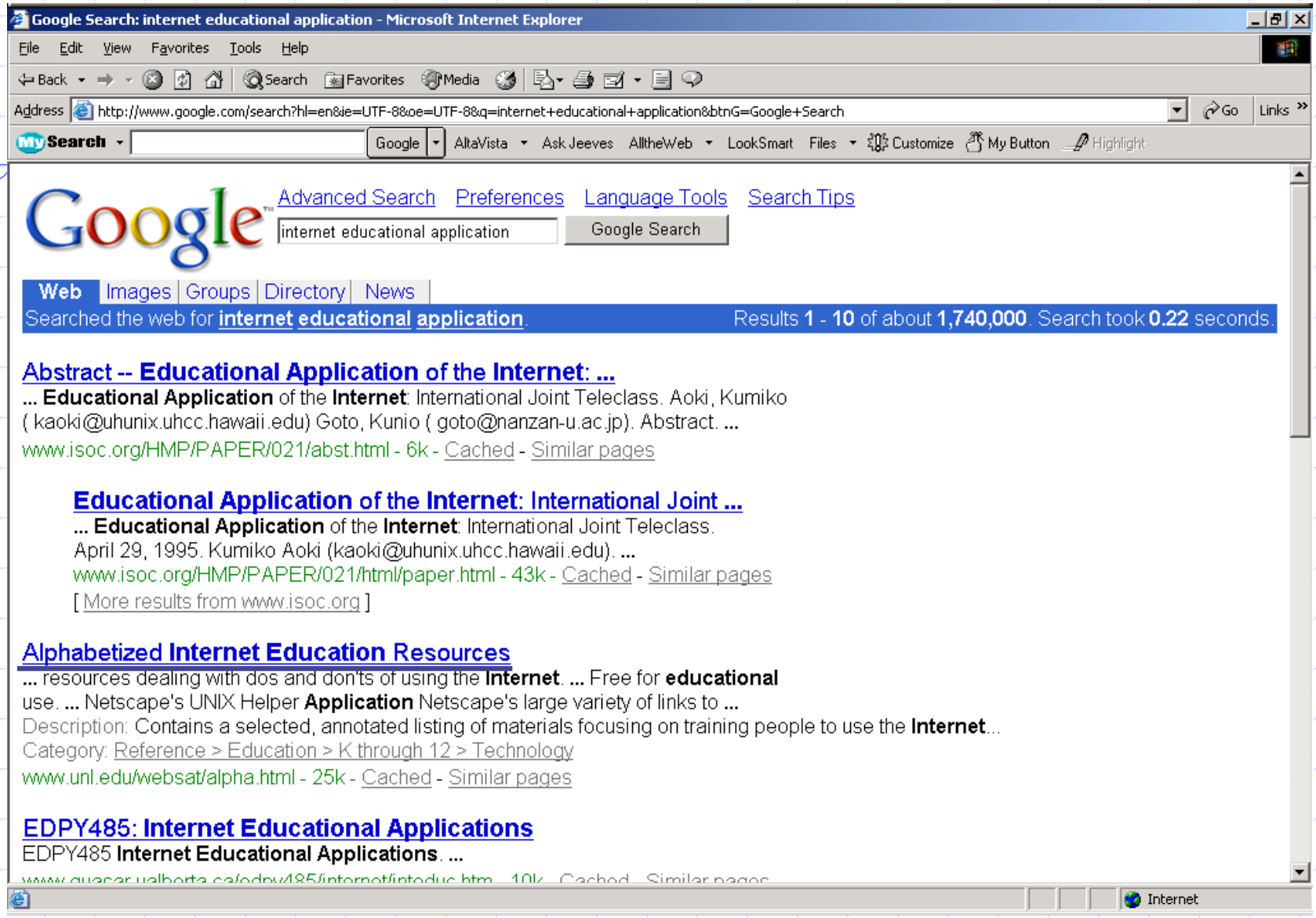
Steps to search for “Internet Educational Applications” using Google:

1. Enter – http://www.google.com
2. Enter – internet educational applications
3. Click on an appropriate link






Google - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Search Favorites Media
Address http://www.google.com
Go Links
MySearch Google AltaVista Ask Jeeves AlltheWeb LookSmart Files Customize My Button Highlight
Google™
Web Images Groups Directory News
Google Search I'm Feeling Lucky
Advanced Search
Preferences
Language Tools
New! The free Google Toolbar blocks pop-ups. Search from anywhere!
Advertise with Us - Business Solutions - Services & Tools - Jobs, Press, & Help - Go to Google Deutschland
Make Google Your Homepage!
©2003 Google - Searching 3,307,998,701 web pages
Done
Internet


Google Search: internet educational application - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Search Favorites Media
Address http://www.google.com/search?hl=en&ie=UTF-8&oe=UTF-8&q=internet+educational+application&btnG=Google+Search
Go Links
MySearch Google AltaVista Ask Jeeves AlltheWeb LookSmart Files Customize My Button Highlight
Google
Advanced Search Preferences Language Tools Search Tips
internet educational application
Google Search
Web Images Groups Directory News
Searched the web for internet educational application.
Results 1 - 10 of about 1,740,000. Search took 0.22 seconds.
Abstract -- Educational Application of the Internet: ...
... Educational Application of the Internet: International Joint Teleclass. Aoki, Kumiko (kaoki@uhunix.uhcc.hawaii.edu) Goto, Kunio (goto@nanzan-u.ac.jp). Abstract. ...
www.isoc.org/HMP/PAPER/021/abst.html - 6k - Cached - Similar pages
Educational Application of the Internet: International Joint ...
... Educational Application of the Internet: International Joint Teleclass. April 29, 1995. Kumiko Aoki (kaoki@uhunix.uhcc.hawaii.edu). ...
www.isoc.org/HMP/PAPER/021/html/paper.html - 43k - Cached - Similar pages
[More results from www.isoc.org]
Alphabetized Internet Education Resources
... resources dealing with dos and don'ts of using the Internet. ... Free for educational use. ... Netscape's UNIX Helper Application Netscape's large variety of links to ...
Description: Contains a selected, annotated listing of materials focusing on training people to use the Internet...
Category: Reference > Education > K through 12 > Technology
www.unl.edu/websat/alpha.html - 25k - Cached - Similar pages
EDPY485: Internet Educational Applications
EDPY485 Internet Educational Applications. ...
Internet

University of Nebraska - Lincoln

Nebraska

UNIVERSITY OF NEBRASKA–LINCOLN

All Resources

This is an alphabetical listing of all resources mentioned by the Internet Education Clearinghouse.

AAHESGIT

Archives of the AAHESGIT listserv, an American Association for Higher Education mailing list about using technology. mailing address of the listserv is AAHESGIT@GWUVM.GWU.EDU.
gopher://info.cren.net/11/archives/aahesgit

American Journal of Distance Education

This journal gives an overview of distance education.
http://www.cde.psu.edu/ACSDE/

Anonymous FTP: Frequently Asked Questions

Information about File Transfer Protocol, including what it is and tips for using it.